

CHAPTER 4 : CURRENT ELECTRICITY AND MAGNETISM

Topic:-

Present all the activities that you performed in this chapter in a Science exhibition.

Introduction:-

Electricity and magnetism are two of the most powerful and fascinating forces in modern science. Today, our modern world cannot function without electric power. From glowing bulbs, fans, and televisions to smartphones and electric trains, electricity powers our daily life. In Chapter 4 of our Class 8 Science book, we study Current Electricity and Magnetism.

Electric current is the continuous flow of negatively charged electrons through a conducting metal wire. In 1820, Danish scientist Hans Christian Oersted made a breakthrough discovery: an electric current flowing through a wire creates a magnetic field around it! This connection between electricity and magnetism is called electromagnetism. In this project, our class built, tested, and demonstrated working electrical models for our annual School Science Exhibition.

❖ Key Activities Covered in Chapter 4:-

- **Activity 1:** Testing simple electrical circuits with dry cell, bulb, and key.
- **Activity 2:** Demonstrating magnetic effect of current using a magnetic compass and copper wire.
- **Activity 3:** Building a DIY working electromagnet using an iron nail and insulated copper wire.
- **Activity 4:** Demonstrating the working mechanism of an electric bell.

Aim & Exhibition Setup:-

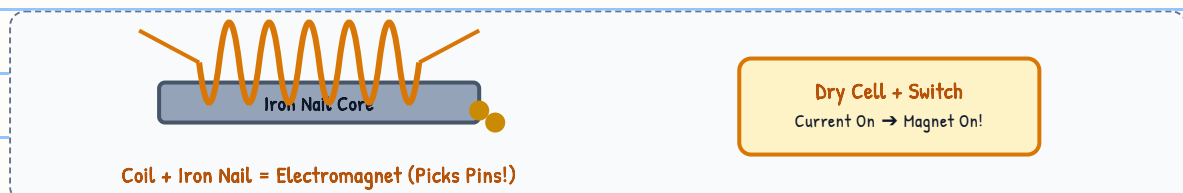
Aim: To build, set up, and present all hands-on electrical and magnetic activities from Chapter 4 in a school science exhibition stall.

❖ *Objectives of the Exhibition Stall:-*

- To demonstrate how electric current flows only in a closed complete circuit.
- To show Oersted's deflection of a magnetic needle when current is switched on.
- To build a working electromagnet that picks up iron pins and paperclips.
- To explain the strike-and-break cycle of an electric door bell.

❖ *Apparatus & Materials Used:-*

- Dry cells (1.5V & 9V batteries), battery holders, connecting copper wires, switches.
- Mini torch bulbs (3V), LED bulbs, magnetic compass needle, large iron nail (10 cm).
- Insulated copper wire (gauge 26), box of iron paper clips, wooden display board.



Procedure & Construction of Exhibits:-

We constructed and mounted four interactive exhibits on a large plywood board:

- **Exhibit 1 (Simple Circuit & Switch):** Connected a dry cell, a plug key switch, and a mini bulb. When the key is plugged in, current flows through the closed loop and the filament glows bright yellow.
- **Exhibit 2 (Oersted's Compass Deflection):** Placed a small magnetic compass needle directly under a straight copper wire. When current was turned on, the magnetic needle deflected instantly! Reversing the battery terminals made the needle deflect in the opposite direction.
- **Exhibit 3 (DIY Electromagnet):** Wrapped 60 tight turns of insulated copper wire around a 10 cm iron nail. Connected the ends to a battery. The nail picked up 15 metal paperclips! When switched off, the clips dropped immediately.
- **Exhibit 4 (Electric Bell Demonstration):** Mounted an electric bell mechanism showing the electromagnet, soft iron armature, striker, gong, and contact screw.

Observations & Exhibition Test Table:-

The table below documents the test results of our four exhibition activities:

<i>Exhibit #</i>	<i>Activity Demonstrated</i>	<i>Key Scientific Observation</i>	<i>Inference / Law Proved</i>
Model 1	Basic circuit with bulb & switch	Bulb glows only when switch is closed	Current requires a complete unbroken circuit
Model 2	Compass needle under wire	Needle deflects 45° when current is ON	Electric current produces a magnetic field
Model 3	Electromagnet with iron nail	Picks up 15 paper clips; drops them when OFF	Temporary magnetism controlled by electric switch
Model 4	Working electric bell model	Striker hits gong continuously (Tring-Tring!)	Make-and-break circuit creates continuous ringing

Scientific Analysis & Discussion:-

Our exhibition models illustrated several fundamental electromagnetic principles:

- **Why the Compass Needle Deflects:** A magnetic compass needle always aligns along Earth's North-South axis. When current flows through the adjacent wire, it creates a circular magnetic field around itself. This new magnetic field exerts a force on the compass needle, pushing it away from North.
- **How Electromagnet Strength Increases:** During live testing, we showed visitors that our nail electromagnet becomes much stronger if:
 - We increase the number of wire coils from 30 turns to 60 turns.
 - We connect two 1.5V batteries in series (higher current flow).
 - We use a soft iron core inside the coil rather than plastic.
- **The Self-Breaking Action in Electric Bells:** When the striker moves forward to hit the gong, its back moves away from the contact screw, breaking the circuit. The magnet turns off, the spring pulls the striker back, contact is remade, and the cycle repeats 20 times a second!

Everyday Applications & Stall Success:-

Electromagnetism is the foundation of almost all modern electrical machines:

- **Electric Motors & Fans:** Use electromagnets to rotate blades and drive washing machines, mixers, and electric cars.
- **Giant Scrap Cranes:** Heavy industrial electromagnets in scrapyards lift tons of heavy iron cars and release them simply by cutting power.
- **Maglev Bullet Trains:** Use powerful electromagnets to levitate trains above tracks, traveling silently at over 500 km/h without wheel friction!
- **Hospital MRI Scanners:** Use superconducting electromagnets to capture detailed internal scans of the human brain and organs.

❖ Judges' Feedback at the Science Exhibition:-

Science exhibition judges praised our interactive stall because visiting students could push the switches and see the compass deflect and the electromagnet lift pins with their own hands! Our stall won the First Prize in the Middle School Physics category.

What We Learned & Conclusion:-

- Electric current flows only through a closed loop conductor powered by a voltage source.
- An electric current creates an invisible magnetic field around itself (Oersted's discovery).
- An electromagnet is a temporary magnet whose magnetic strength can be turned on, off, and regulated easily.
- Electric bells, motors, and scrap cranes operate on the magnetic effects of electric current.

Conclusion:-

Presenting all the activities from Chapter 4 in our school science exhibition was a wonderful hands-on learning adventure. Building working circuits, Oersted deflections, electromagnets, and electric bells turned theoretical textbook physics into practical, visible machines. The project ignited our curiosity to explore electronics and electrical engineering in higher classes.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 4: Current Electricity and Magnetism).
- Experiments in Electricity & Magnetism, Balbharti Pune.
- Science Club Hands-On Demonstration Manual for Class 8.